



American International University-Bangladesh (AIUB) Department of Computer Science

Course Name: **COMPUTER GRAPHICS**
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Course Code: CSE 4118
Semester: **FALL 2025-2026**
MID Term LAB Task-4
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Object & Midpoint Circle

1. Using opengl Implement the Midpoint Circle Algorithm where the center is $(-2, -2)$ and the radius is 50. The range of X-Axis is -200 to 200 and Y-Axis is -200 and 200.
2. Draw the following:



OpenGL Code

Question - 1

Midpoint Circle

```
#include<windows.h>
#include<iostream>
#include<math.h>
#include <stdio.h>
#include<GL/gl.h>
#include <GL/glut.h>

using namespace std;

int xc = -2; // center x
int yc = -2; // center y
int r = 50; // radius

// Function to draw 8 symmetric points
void draw8Points(int xc, int yc, int x, int y)
{
    glBegin(GL_POINTS);
        glVertex2i(xc + x, yc + y);
        glVertex2i(xc - x, yc + y);
        glVertex2i(xc + x, yc - y);
        glVertex2i(xc - x, yc - y);

        glVertex2i(xc + y, yc + x);
        glVertex2i(xc - y, yc + x);
        glVertex2i(xc + y, yc - x);
        glVertex2i(xc - y, yc - x);
    glEnd();
}

// Midpoint Circle Algorithm
void midpointCircle()
```

```

{
    int x = 0;
    int y = r;
    int d = 1 - r; // initial decision parameter

    draw8Points(xc, yc, x, y);

    while (x < y)
    {
        if (d < 0)
        {
            d += 2 * x + 3;
        }
        else
        {
            d += 2 * (x - y) + 5;
            y--;
        }
        x++;

        draw8Points(xc, yc, x, y);
    }
}

void display(void)
{
    glClear(GL_COLOR_BUFFER_BIT);
    midpointCircle();
    glFlush();
}

void myInit(void)
{
    glClearColor(1, 1, 1, 0);
    glColor3f(1, 0, 0);

```

```
glPointSize(3.0);

glMatrixMode(GL_PROJECTION);
glLoadIdentity();
// Required coordinate system
gluOrtho2D(-200, 200, -200, 200);
}

int main(int argc, char** argv)
{
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(640, 480);
    glutInitWindowPosition(100, 150);
    glutCreateWindow("Midpoint Circle Algorithm");

    glutDisplayFunc(display);
    myInit();
    glutMainLoop();

    return 0;
}
```

Output



Question - 2

Object: Bangladesh Flag

```
#include<windows.h>

#include<iostream>

#include<math.h>

#include <stdio.h>

#include<GL/gl.h>

#include <GL/glut.h>


using namespace std;


float PI = 3.1416;


// Draw a Filled Circle using Triangle Fan
void drawFilledCircle(GLfloat x, GLfloat y, GLfloat radius)
{
    int triangleAmount = 100;

    float twicePi = 2 * PI;

    glBegin(GL_TRIANGLE_FAN);
        glVertex2f(x, y); // circle center
        for(int i = 0; i <= triangleAmount; i++)
        {
            glVertex2f
            (
                x + (radius * cos(i * twicePi / triangleAmount)),
                y + (radius * sin(i * twicePi / triangleAmount))
            );
        }
    glEnd();
}


void display(void)
{

```

```

glClear(GL_COLOR_BUFFER_BIT);

// Center offset

int cx = 50;
int cy = 20;


// FLAG POLE - BLACK

glColor3ub(0,0,0);
glLineWidth(6.0);
glBegin(GL_LINES);
    glVertex2i(-150 + cx, -150 + cy);
    glVertex2i(-150 + cx, 70 + cy);
glEnd();


// Pole tip triangle - BLACK

glBegin(GL_TRIANGLES);
    glColor3ub(0,0,0);
    glVertex2i(-150 + cx, 75 + cy);
    glVertex2i(-155 + cx, 62 + cy);
    glVertex2i(-145 + cx, 62 + cy);
glEnd();


// GREEN RECTANGULAR

glColor3ub(0, 106, 78);
glBegin(GL_QUADS);
    glVertex2i(-150 + cx, 60 + cy);
    glVertex2i(-50 + cx, 60 + cy);
    glVertex2i(-50 + cx, -60 + cy);
    glVertex2i(-150 + cx, -60 + cy);
glEnd();


// RED CIRCLE

glColor3ub(244, 42, 65);
drawFilledCircle(-50 + cx, 0 + cy, 35);

```

```

// WHITE STRIP ON POLE

glColor3ub(255, 255, 255);

glLineWidth(7.0);

glBegin(GL_LINES);

    glVertex2i(-150 + cx, -60 + cy);

    glVertex2i(-150 + cx, 60 + cy);

glEnd();

glFlush();
}

// INITIALIZATION FUNCTION

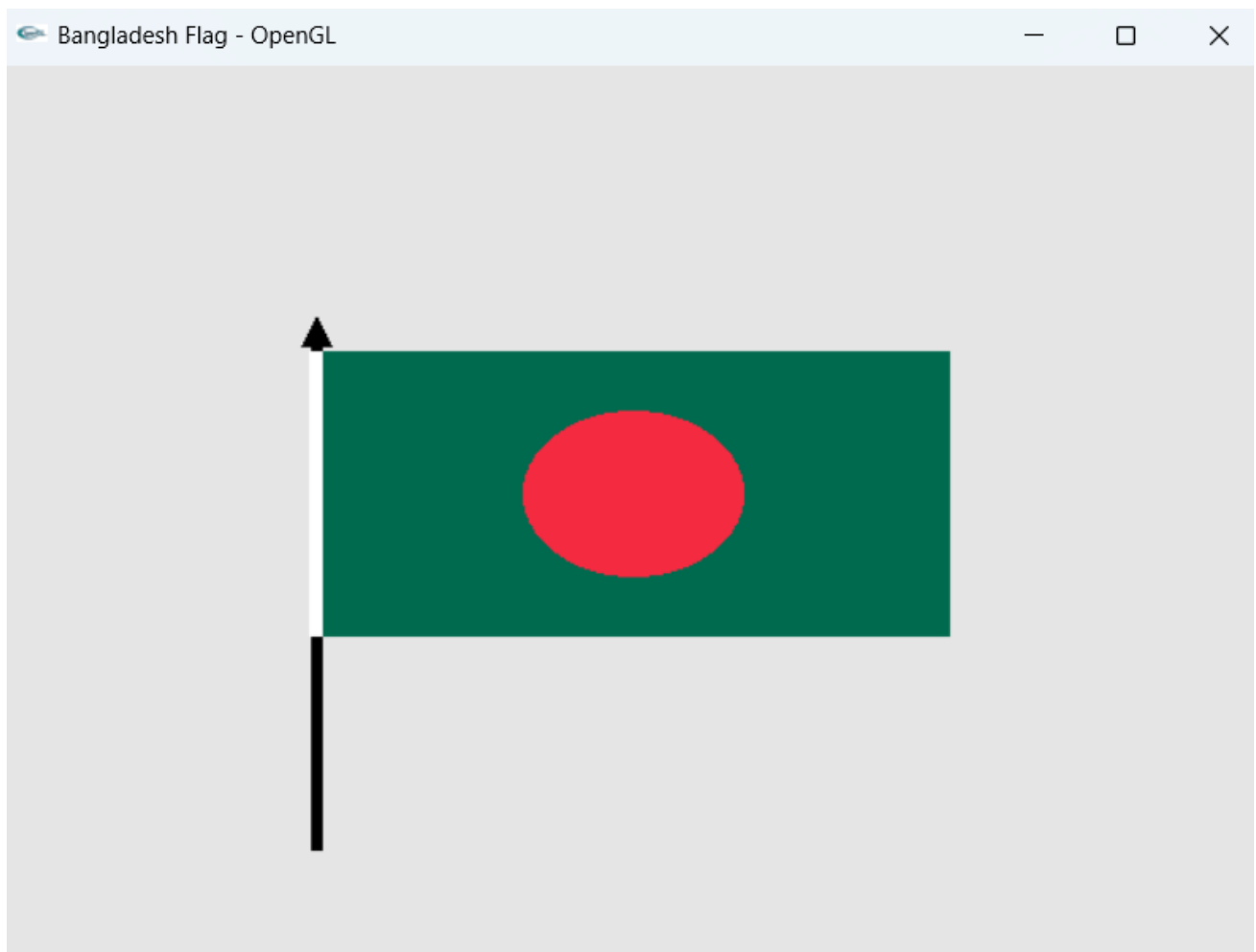
void myInit(void)
{
    glClearColor(0.90, 0.90, 0.90, 0.0); // Light gray background
    glColor3f(0,0,0);
    glPointSize(3.0);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(-200, 200, -200, 200);
}

int main(int argc, char** argv)
{
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(640, 480);
    glutInitWindowPosition(100, 150);
    glutCreateWindow("Bangladesh Flag - OpenGL");
    glutDisplayFunc(display);
    myInit();
    glutMainLoop();

    return 0;
}

```

Output



THE END